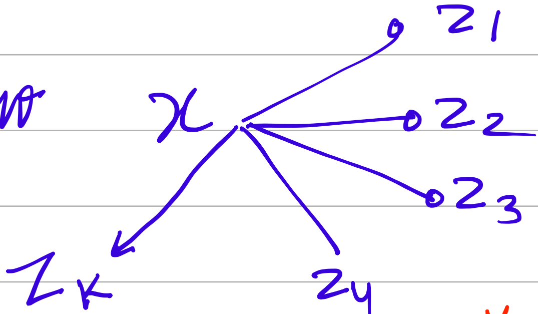
(1) $p_\theta(x) \Rightarrow$ This is probability of x belonging to the distribution which is parametrized by θ

basically $\{x \in \mathcal{D}\}$

Our assumption is gaussian distribution
$\theta \Rightarrow \mu_\theta$ & σ_θ

(2) $l(\theta) = \log p_\theta(x)$

now



$\therefore p(x) = p(x, z_1) + p(x, z_2) + \ldots p(x, z_K)$

$$p(x) = \sum_{i=1}^{K} p(x, z_i)$$

$$\therefore l(\theta) = \log \sum_{z} p_\theta(x, z)$$

basically each $x \Rightarrow z_i$
Associated to

Called as x marginalized over all z

(3) $$l(\theta) = \log \sum_{z} p_\theta(x,z) * \frac{q(z|x)}{q(z|x)}$$

$$= \log \sum_{z} q(z|x) * \frac{p_\theta(x,z)}{q(z|x)}$$

$\because$ $\mathbb{E}(x) = \int x\, p(x)\, dx$

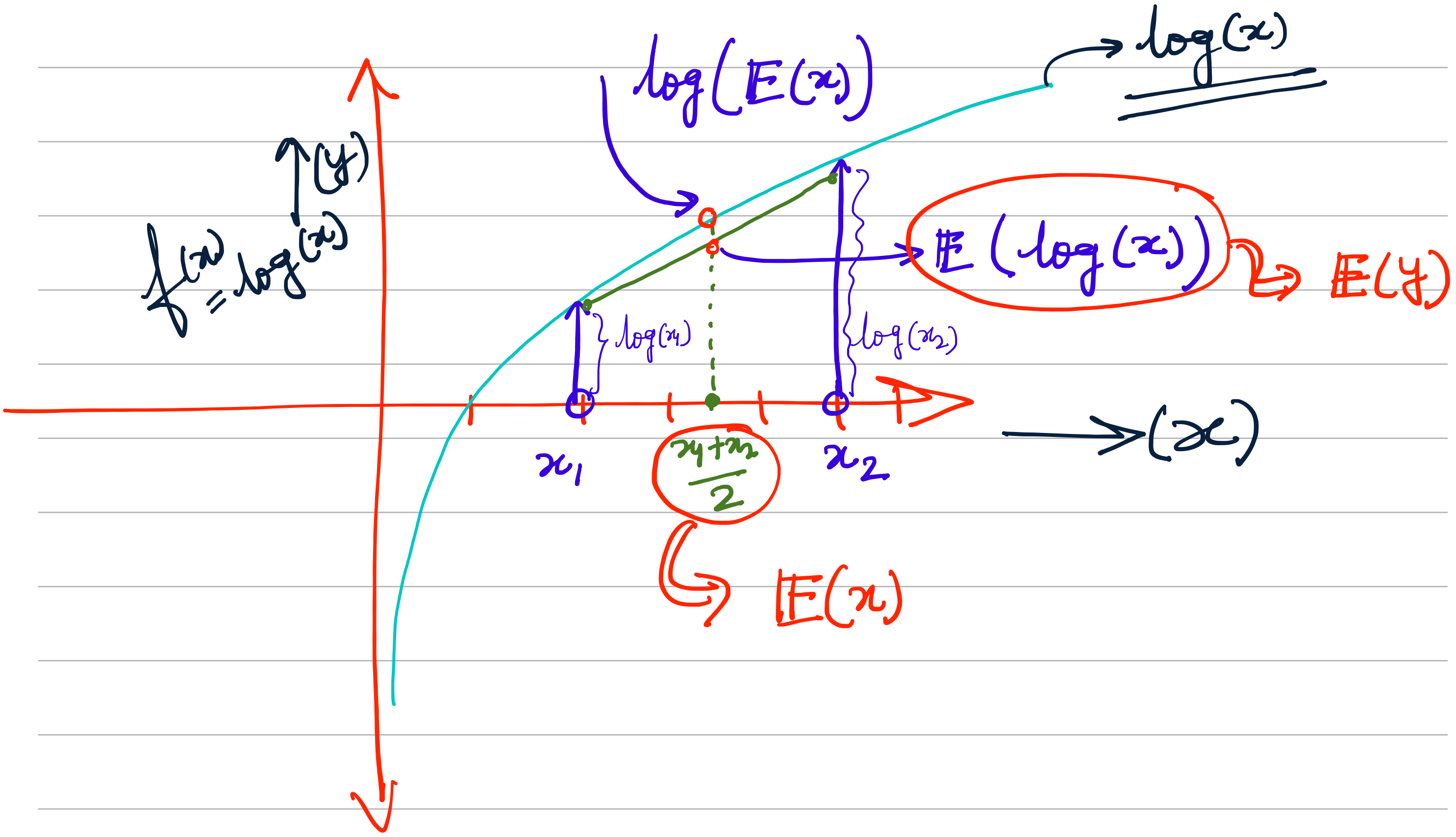
(4) $$l(\theta) = \log \underbrace{\mathbb{E}}_{\underbrace{q(z|x)}_{B}} \left[\underbrace{\frac{p_\theta(x,z)}{q(z|x)}}_{A} \right]$$

basically expectation of (A) wrt (B)

(5) Jensen inequality suggests that

log of expectation is > expectation of log

$\log(\mathbb{E}[\ ]) > \mathbb{E}(\log[\ ])$



$$\therefore \log \mathbb{E}_{q(z|x)}\left[\frac{p_\theta(x,z)}{q(z|x)}\right] \geq \mathbb{E}_{q(z|x)}\left[\log \frac{p_\theta(x,z)}{q(z|x)}\right]$$

(6) $$\ell(\theta) \geq \mathbb{E}_{q(z|x)}\left(\log\left[\frac{p_\theta(x,z)}{q(z|x)}\right]\right) = F_\theta(q)$$

[ELBO]

Evidence Lower Bound.

(7) $$F_\theta(q) = \mathbb{E}_{q(z|x)} \log\left[\frac{p_\theta(x|z)\, p_\theta(z)}{q(z|x)}\right]$$

(Bayes rule)

(8) $$F_\theta(q) = \mathbb{E}_{q(z|x)}\left[\log p_\theta(x|z)\right] + \mathbb{E}_{q(z|x)}\left[\log \frac{p_\theta(z)}{q(z|x)}\right]$$

$$= \mathbb{E}_{q(z|x)}\left[\log p_\theta(x|z)\right] - \mathbb{E}_{q(z|x)}\left[\log\left(\frac{q(z|x)}{p_\theta(z)}\right)\right]$$

(9) $$F_\theta(q) = \mathbb{E}_{q(z|x)}\left[\log p_\theta(x|z)\right] - \sum q(z|x) * \log\left(\frac{q(z|x)}{p_\theta(z)}\right)$$

By def. of Exp

$A \log\left(\frac{A}{B}\right)$

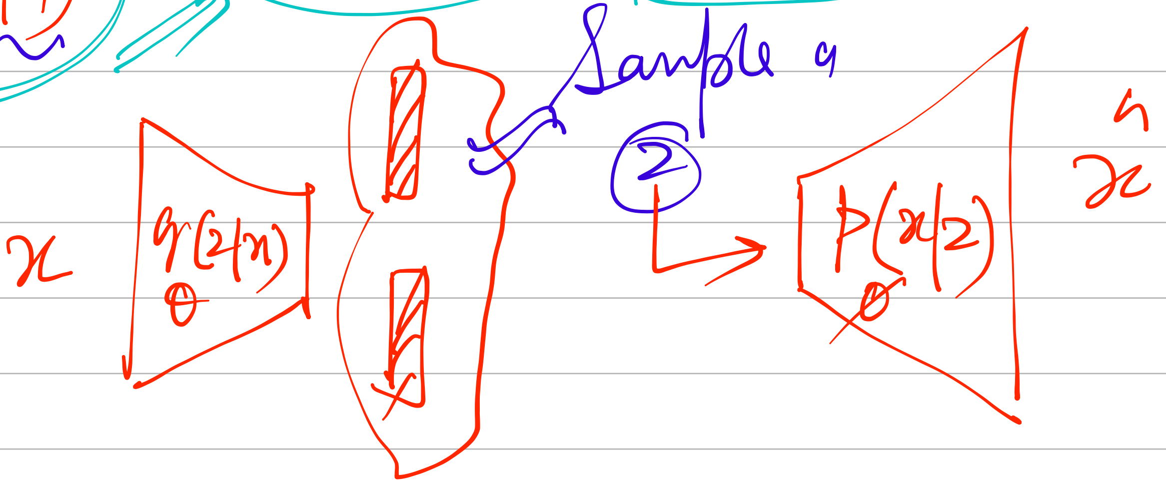
$\underbrace{A \log A}_{} - \underbrace{A \log B}_{}$

(10) $$F_\theta(q) = \mathbb{E}_{q_\phi(z|x)}\left[\log p_\theta(x|z)\right] - D_{KL}\Big(\underbrace{q_\phi(z|x)}_{\text{Posterior}} \,\Big|\, \underbrace{p_\theta(z)}_{\text{Prior}}\Big)$$

Known fixed $\mathcal{N}(0,1)$

$$E_{q(z|x)} \log p(x|z) - D_{KL}\left(q(z|x) \,|\, N(0,1)\right)$$

$$p(x|z) \approx e^{-(x)^2}$$

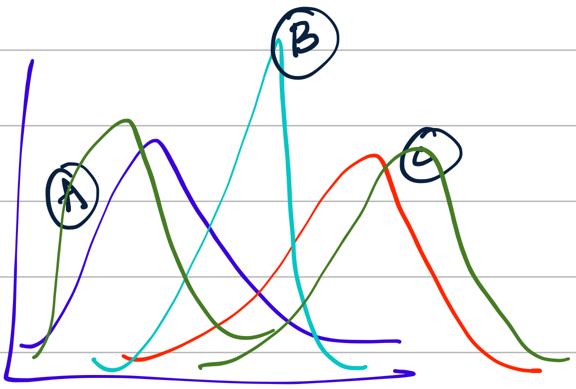


X (Event)	P(X)	-log(P(X))
Virat Scored Century	↑	↓
Aman Scored Century	↓	↑

$$H = \sum_x -p(x)\log(p(x))$$

$\mathbb{E}$ (value of information) $\Rightarrow$ Entropy

$\circledast$



p | q

Entropy(q) − Ent(p)

$$-\sum q(x)\log(q(x)) + \sum p(x)\log p(x)$$

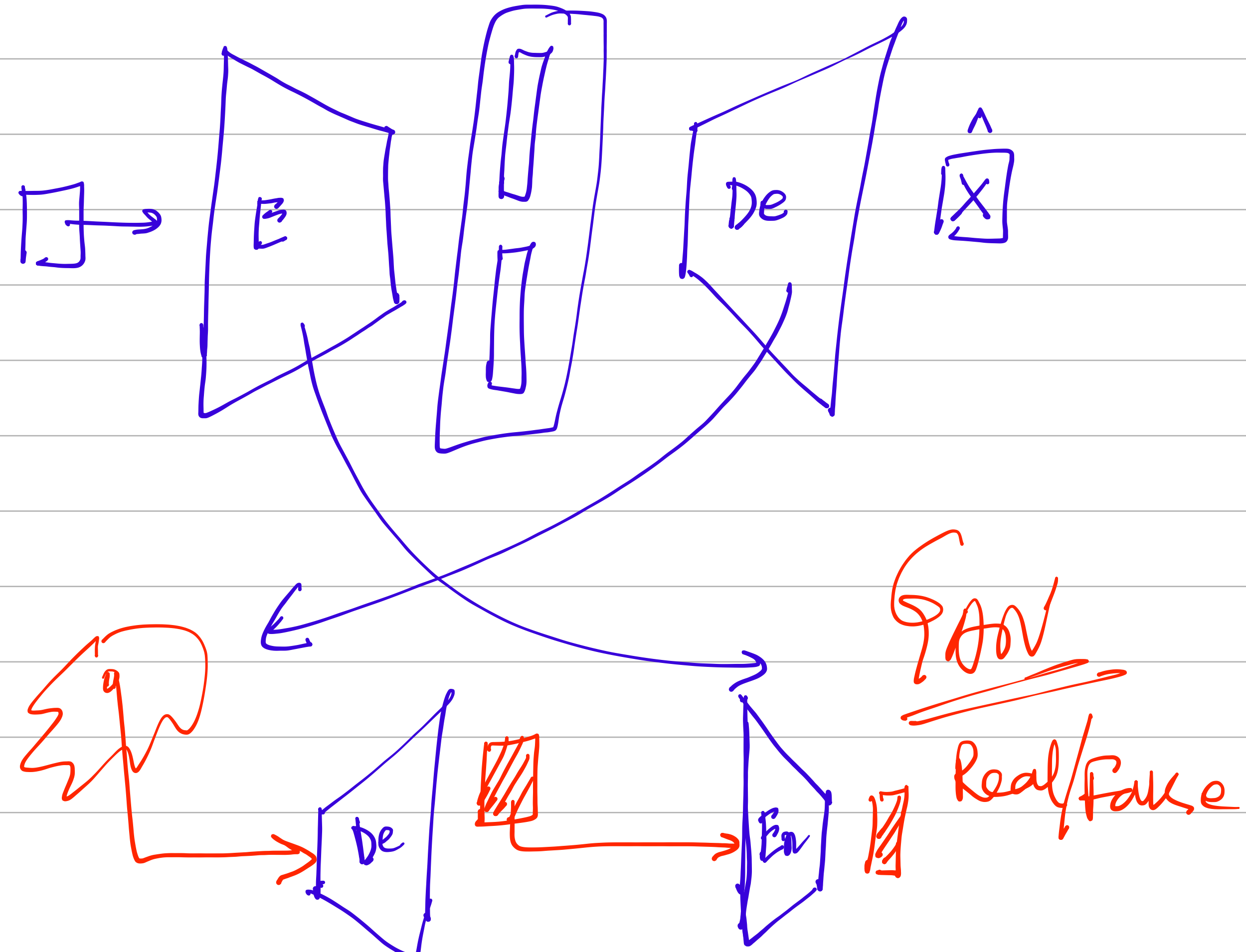
$$\Downarrow$$

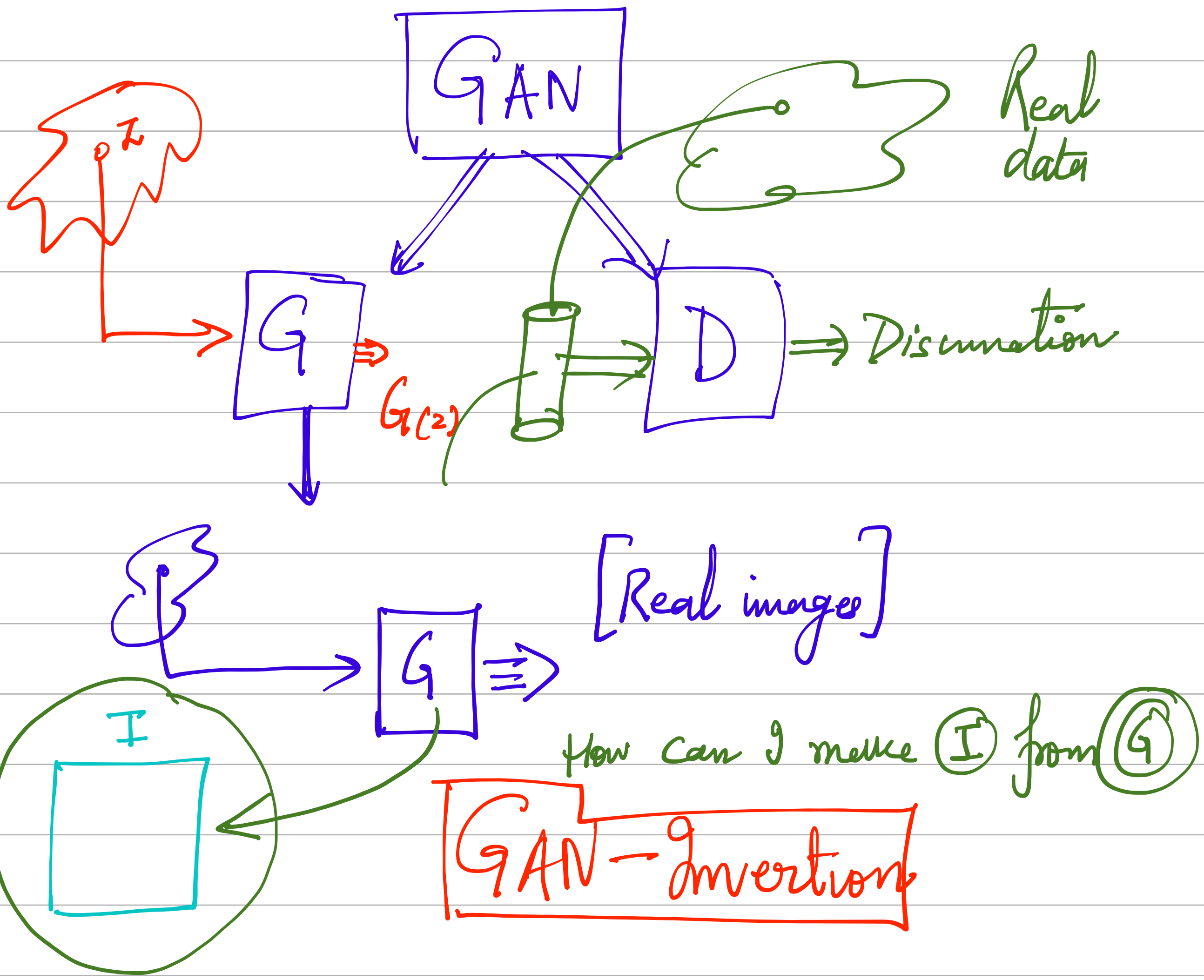
$$\underbrace{-\sum p(x)\log(q(x))}_{CE} + \underbrace{\sum p(x)\log p(x)}_{En}$$

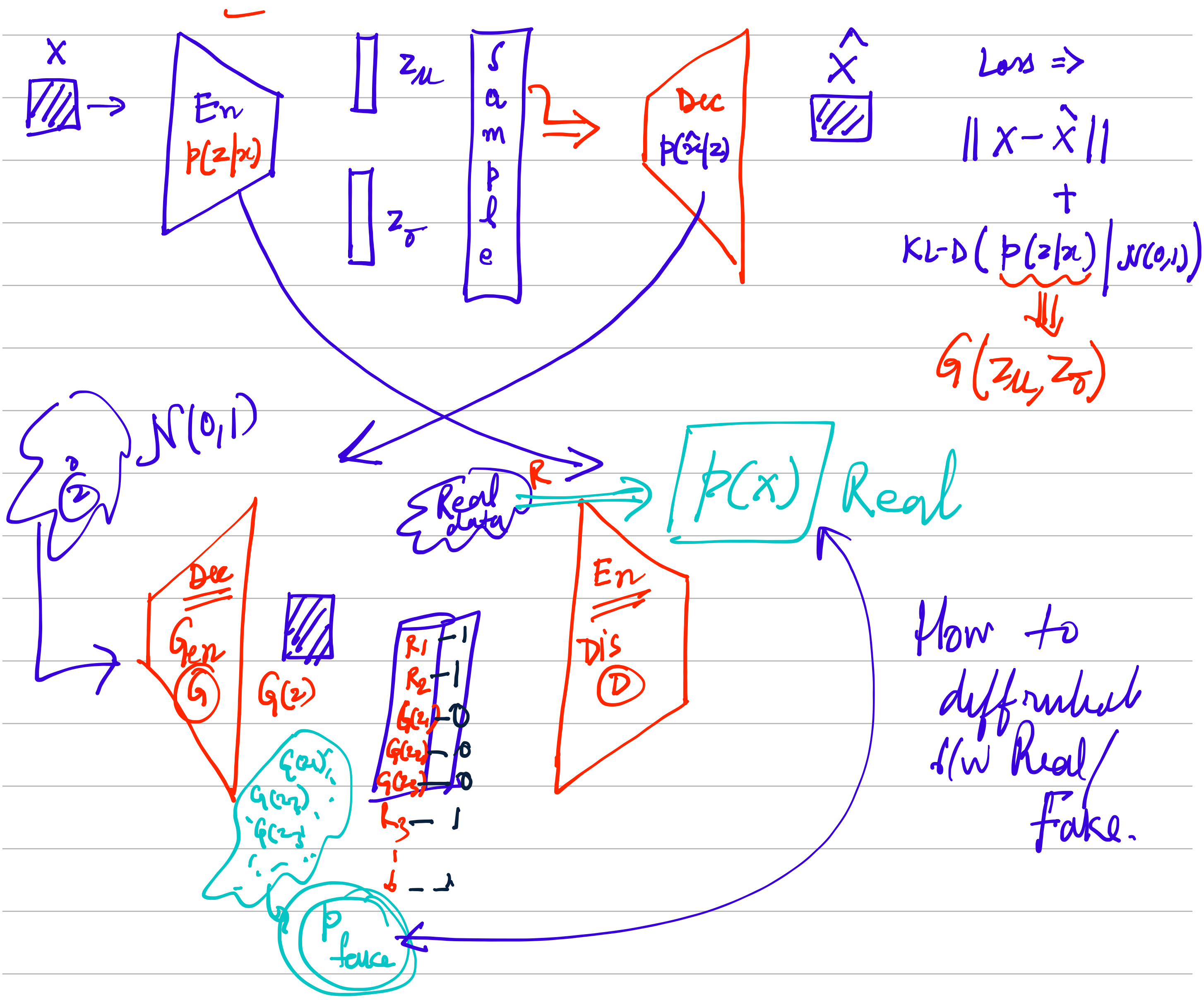
$$\text{KL-Div}(p(x) \| q(x)) = -\sum p(x)\log q(x) + \sum p(x)\log p(x)$$

$$= \sum p(x)\log\frac{p(x)}{q(x)}$$

$$\Rightarrow -\sum p(x) \log\left[\frac{q(x)}{p(x)}\right]$$




GAN
Real data
z
G
D
$G(z)$
Discrimination
Real images
I
How can I make I from G
GAN-Inversion


X
En
p(z|x)
z_μ
z_σ
Sample
Dec
$p(\hat{x}|z)$
$\hat{X}$
Loss =>
$\|X - \hat{X}\|$
+
KL-D(p(z|x) | N(0,1))
G(z_μ, z_σ)
N(0,1)
z
Dec
Gen
G
G(z)
Real data
R
p(x)
Real
En
Dis
D
R_1 - 1
R_2 - 1
$G(z_1)$ - 0
$G(z_2)$ - 0
$G(z_3)$ - 0
R_3 - 1
$G(z_1)$
$G(z_2)$
$G(z_3)$
p_{fake}
How to differentiate b/w Real/Fake.

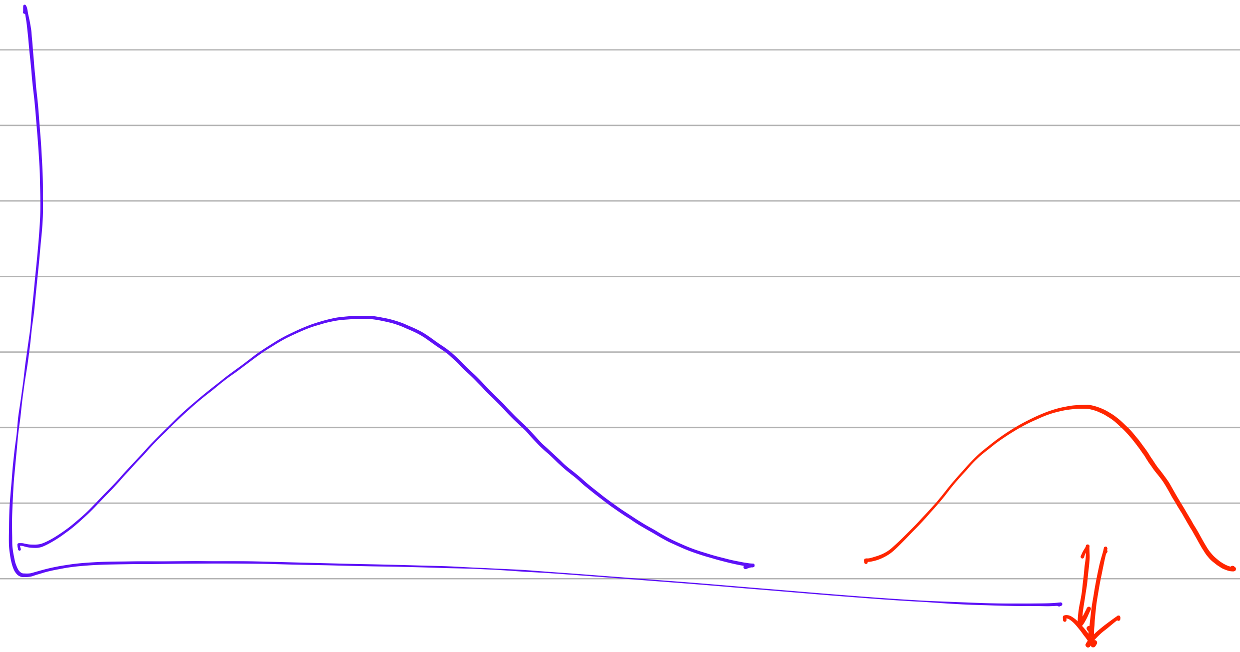
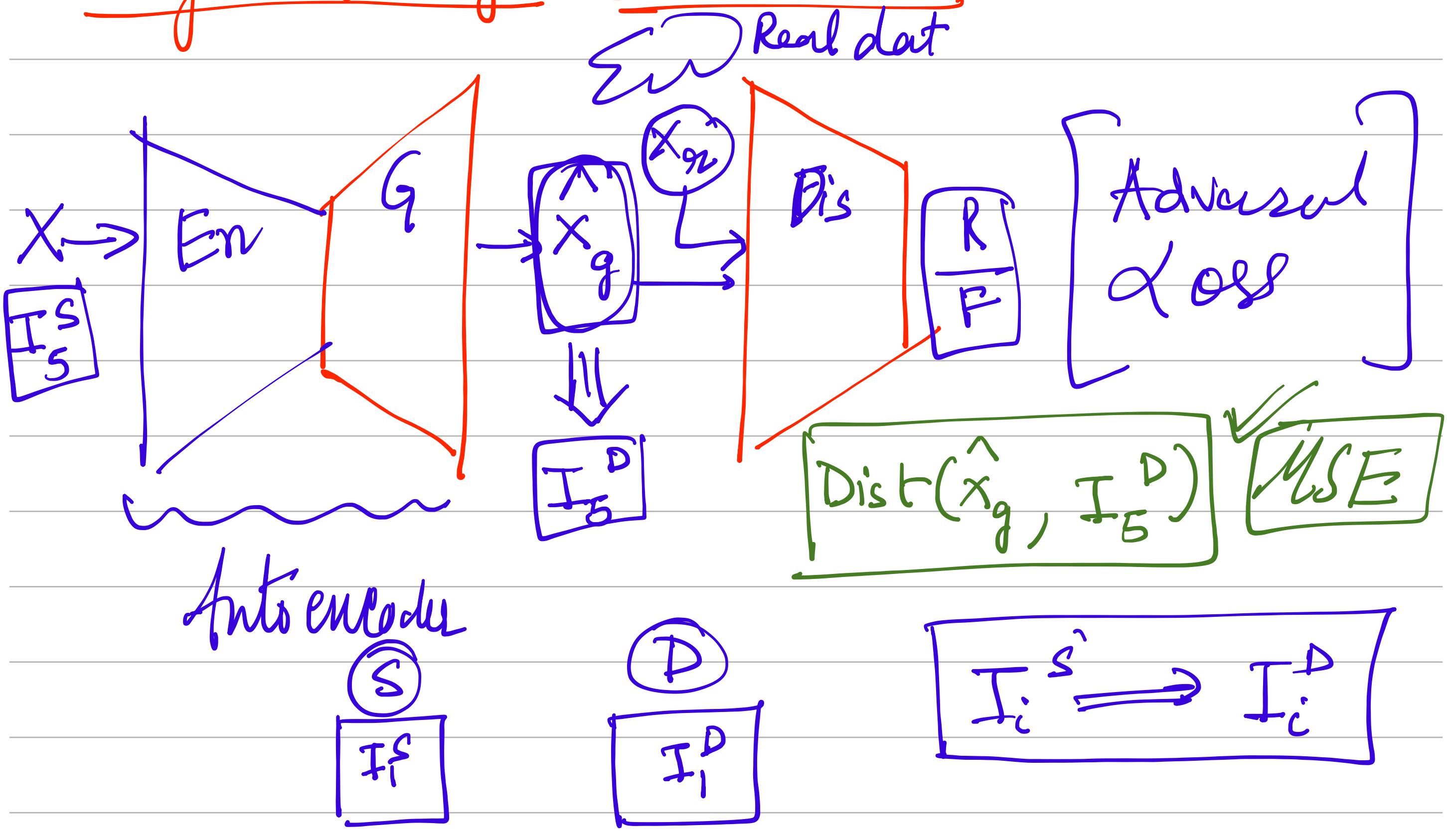
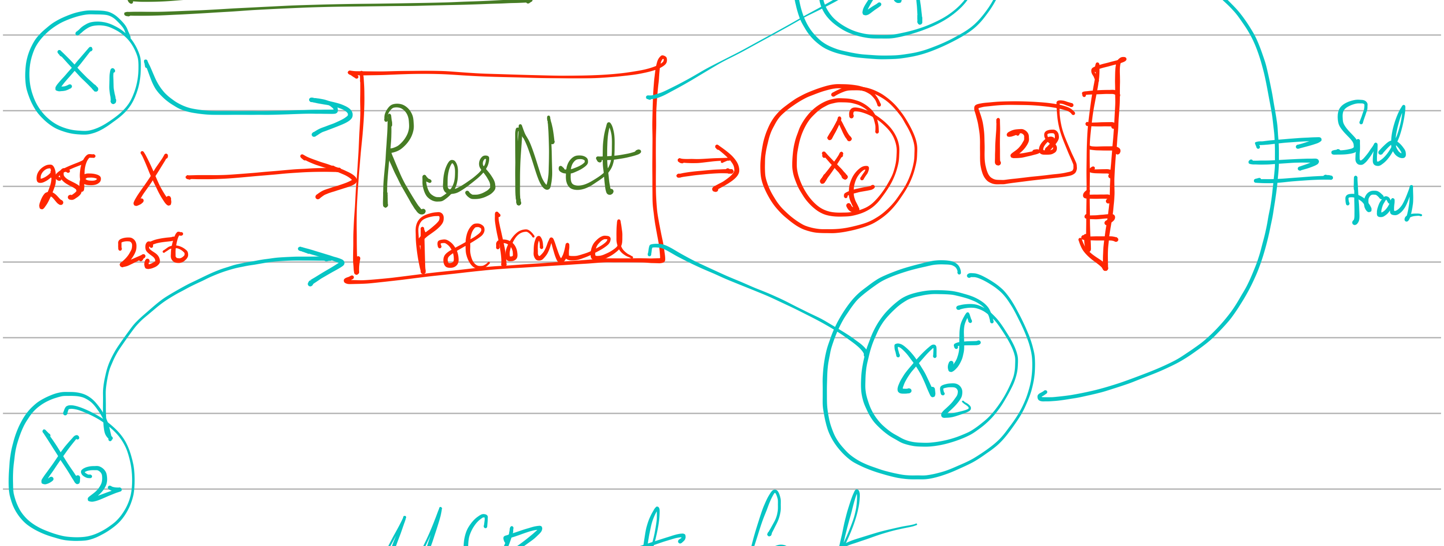


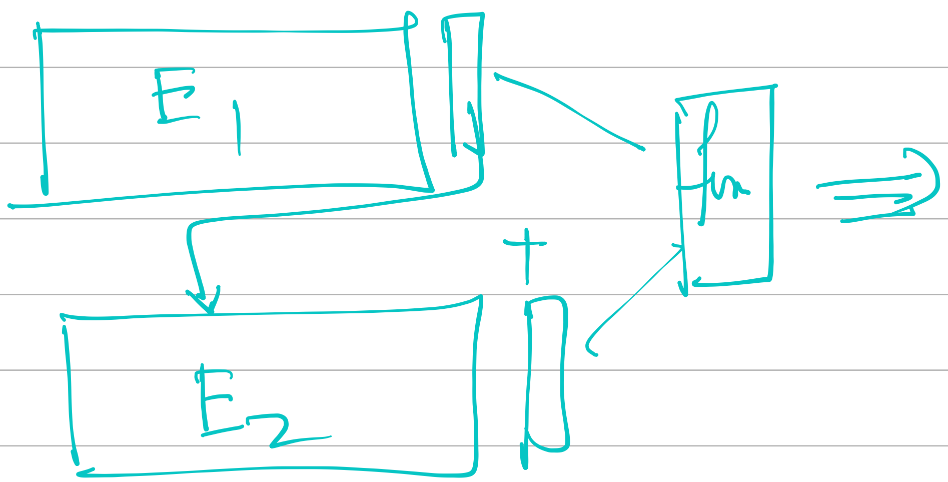
Image 2 Image Translation (Paired)



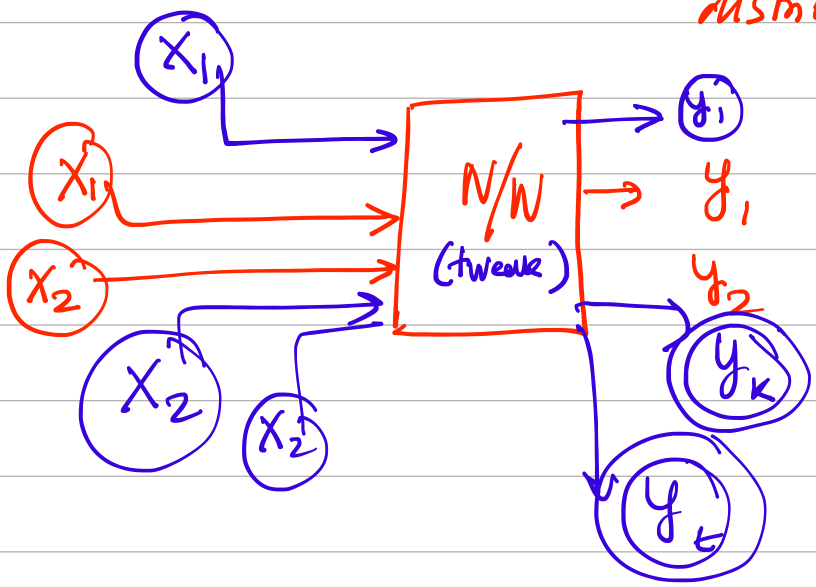
Content Loss

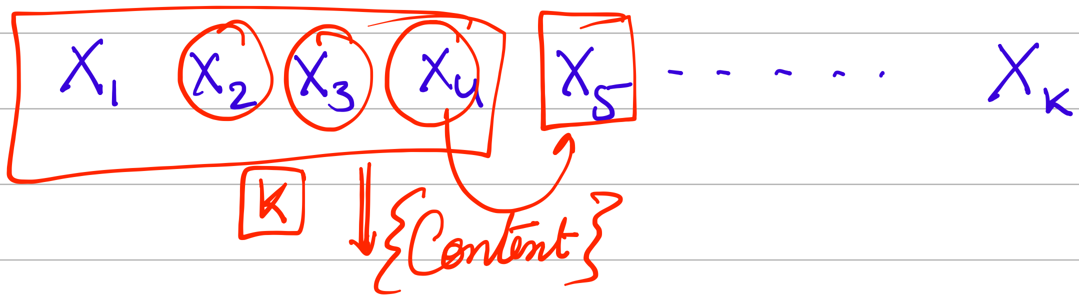


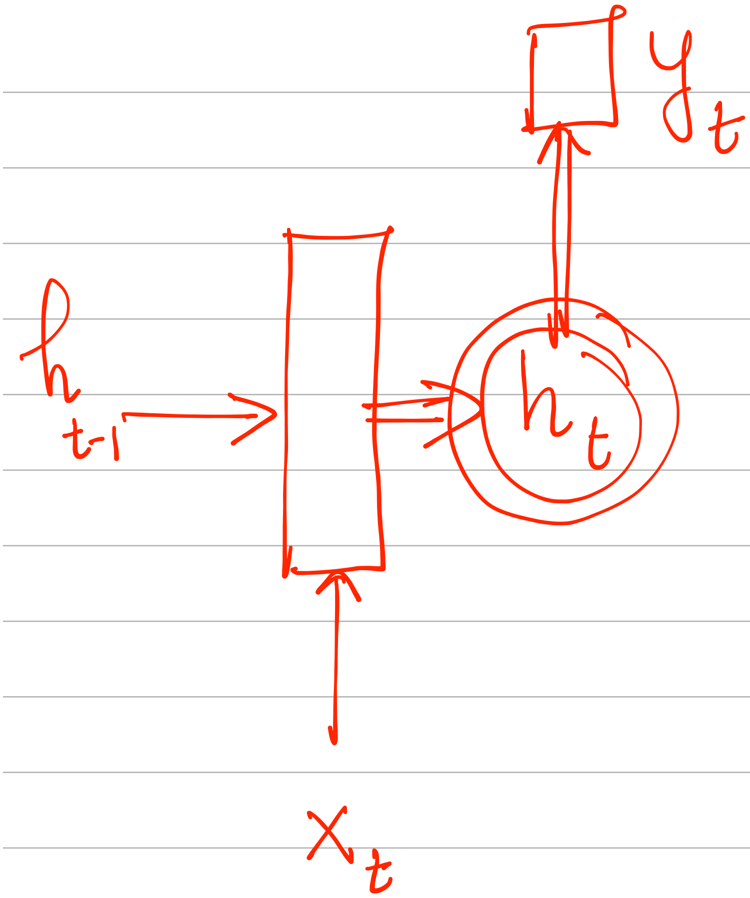
MSE at feature Level



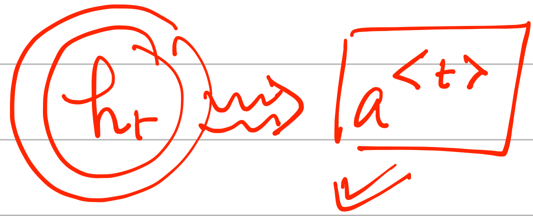
$\boxed{\text{IID}}$ (Identically Independently distributed)







$$a^{<0>} = \vec{0}$$



$$a^{<t>} = g_1(W_{aa} a^{<t-1>} + W_{ax} x^{<t>} + b_a)$$

$$\hat{y}^{<t>} = g_2(W_{ya} a^{<t>} + b_y)$$

$$\underbrace{W_{aa}}_{1000 \times 1000} \underbrace{a^{<t-1>}}_{1000 \times 1} + \underbrace{W_{ax}}_{1000 \times 100} \underbrace{x^{<t>}}_{100 \times 1} + \underbrace{b_a}_{1000 \times 1}$$

$\Rightarrow 1000 \times 1100$

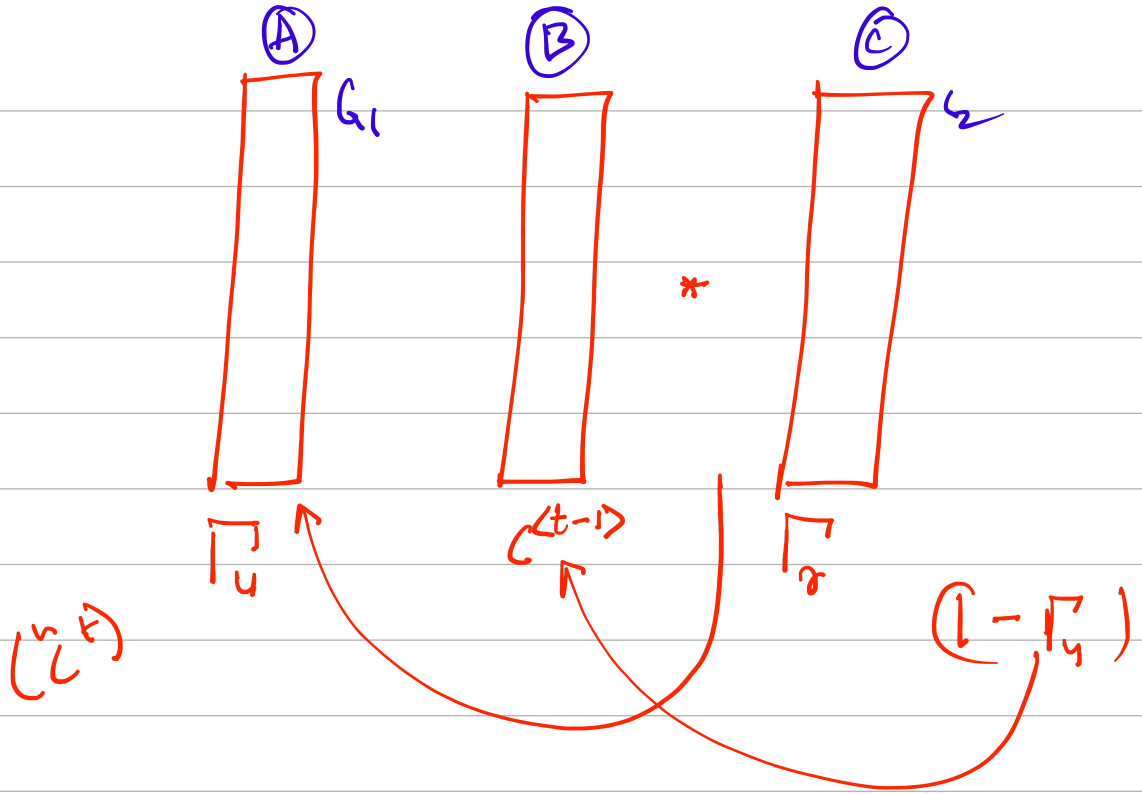
$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} \left.\begin{matrix} 1 \\ 2 \\ 3 \end{matrix}\right\} a \\ \left.\begin{matrix} 4 \\ 5 \\ 6 \end{matrix}\right\} b \end{bmatrix}$$

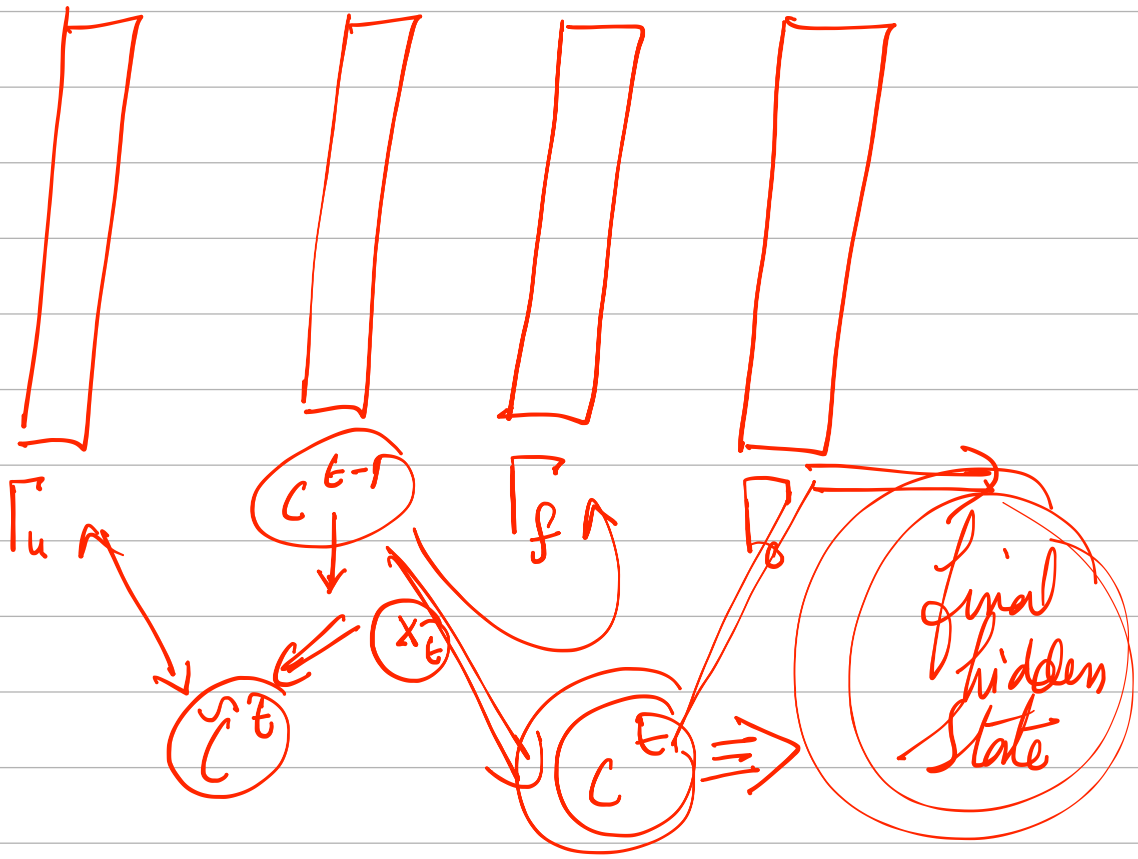
3×6

(6×1)

3×1

$$\Rightarrow \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix} \quad \boxed{a+b}$$


A
B
C
G_1
G_2
*
Γ_u
$c^{<t-1>}$
Γ_r
$(1 - \Gamma_u)$
$\tilde{c}^{<t>}$


Γ_u
c^{t-1}
Γ_f
Γ_o
x_t
$\tilde{c}^{t}$
c^{t}
final hidden state

100 ep ⇒ 95%
7days

LM

100 GB

Transfors 2day

80% 100ep

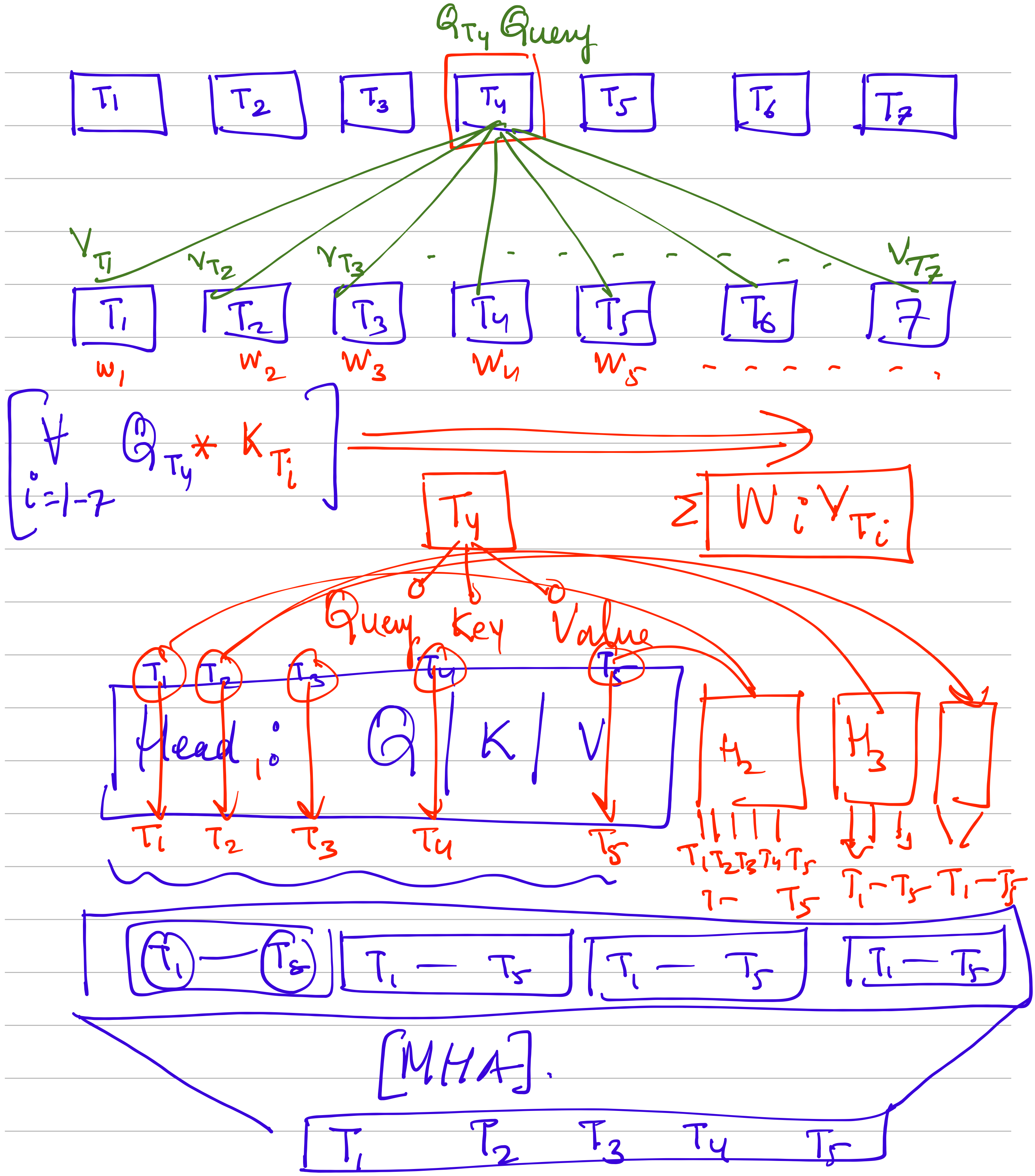
LSTM

90%

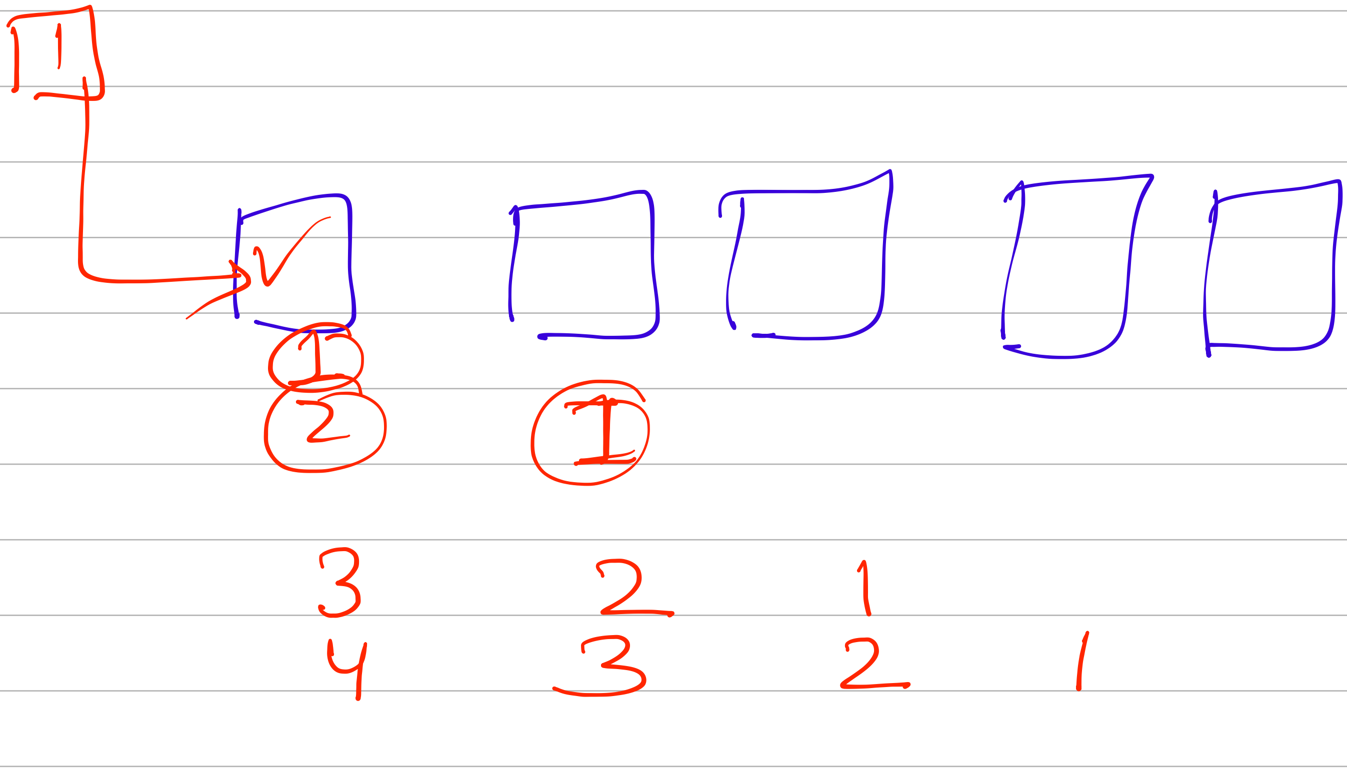
LLM

1000 GB

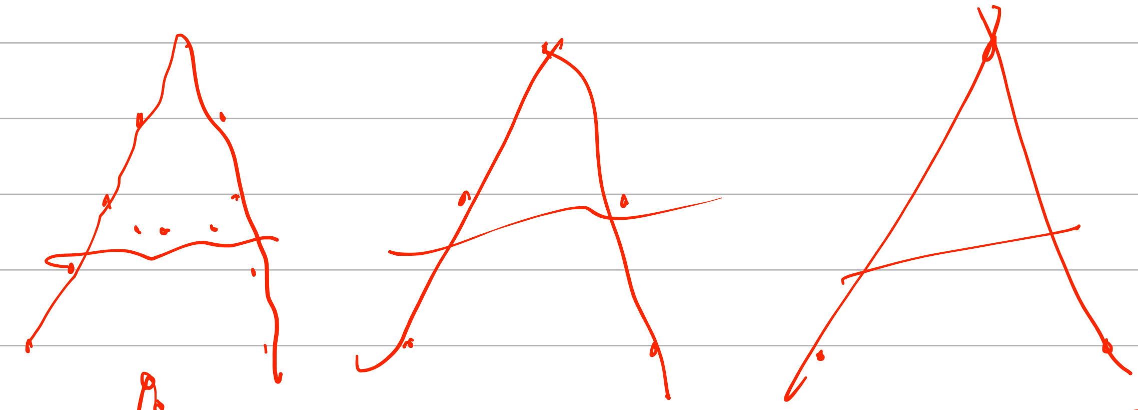
~~LSTM~~ X



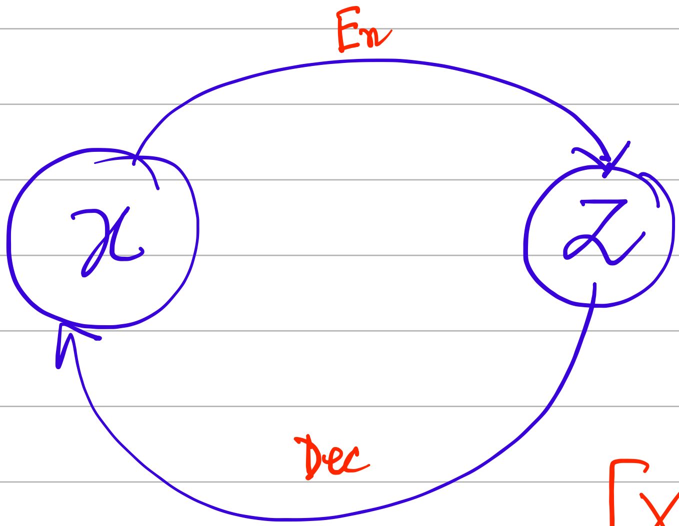
Q_T4 Query
T1 T2 T3 T4 T5 T6 T7
V_T1 V_T2 V_T3 V_T7
T1 T2 T3 T4 T5 T6 7
w1 w2 w3 w4 w5
∀ i=1-7 [Q_T4 * K_Ti]
T4
Σ W_i V_Ti
Query Key Value
T1 T2 T3 T4 T5
Head i : Q | K | V
H2 H3
T1 T2 T3 T4 T5
T1 T2 T3 T4 T5
T1 – T5
T1 – T5 T1 – T5
T1 — T5 T1 — T5 T1 — T5 T1 — T5
[MHA].
T1 T2 T3 T4 T5



1
1
2
1
3 2 1
4 3 2 1



Stuned of my data

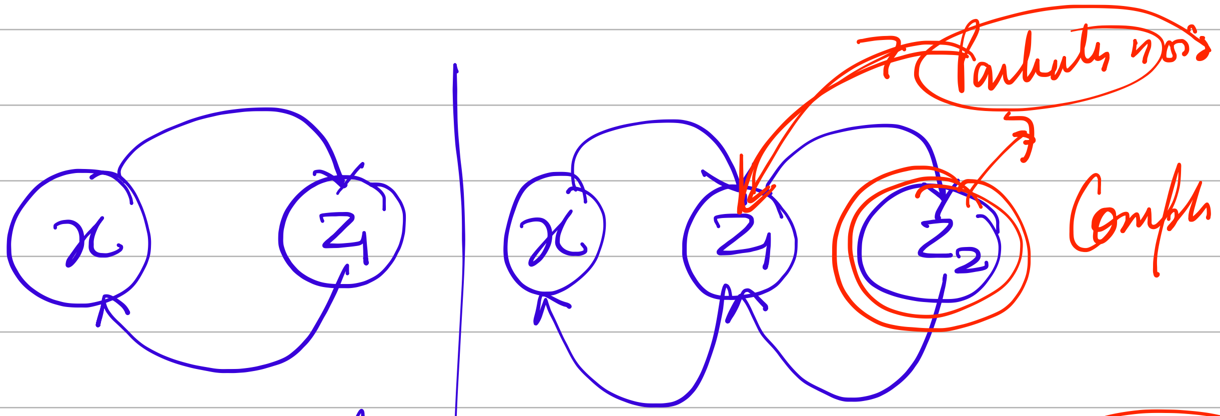


(VAE)

happned in 1 shot.

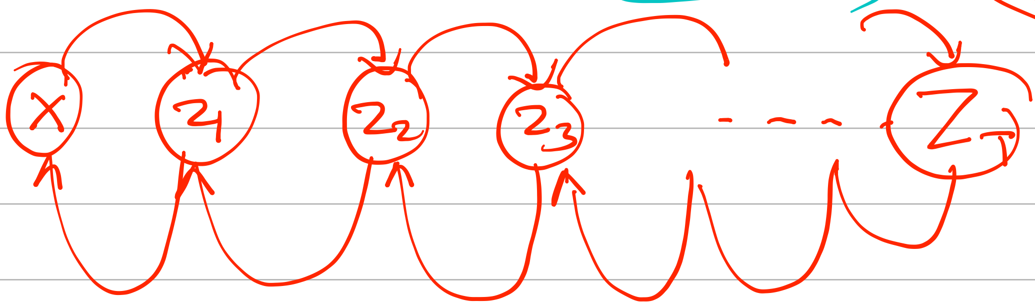
$[X \Longrightarrow Z]$

$z \equiv z \sim \mathcal{N}(0,1)$ Random noise



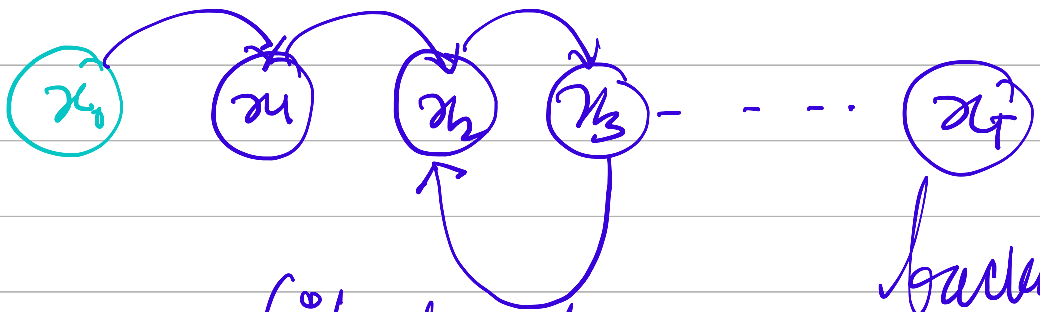
Learning forward

HVAE



Backward

forward



(iteration denoising)

backward

x_0

$x_1 = x_0 + \epsilon_1$

$x_2 = x_1 + \epsilon_2 = x_0 + \epsilon_1 + \epsilon_2$

$x_3 = x_2 + \epsilon_3$

$$x_t = x_{t-1} + \left[\underbrace{\epsilon_1 + \epsilon_2 + \epsilon_3 \cdots\cdots \epsilon_t}\right]$$

$\epsilon_i \in \mathcal{N}(0,1)$

$\Downarrow$

ϵ [Big μ & σ]

$$\begin{tikzcd} x_{t-1} \arrow[r, bend left, "f"] & x_t \arrow[l, bend left, "b"] \end{tikzcd}$$

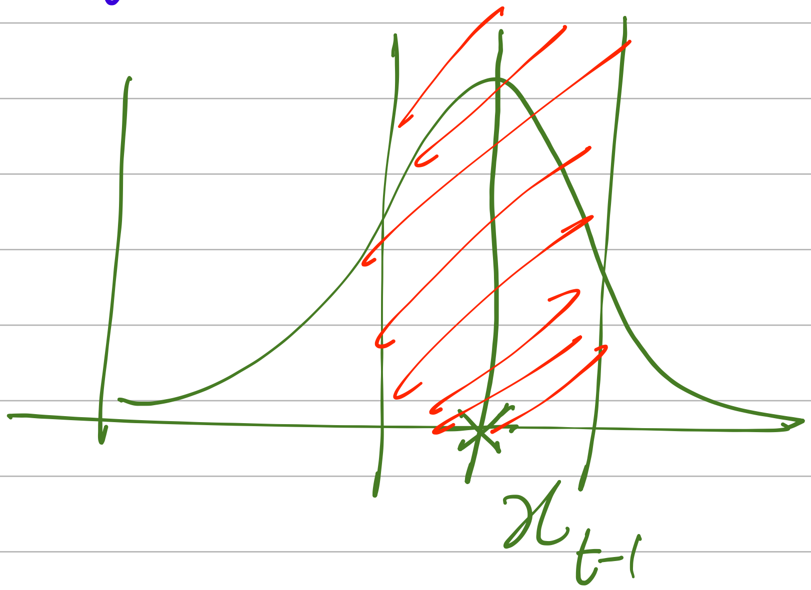
$q(x_t | x_{t-1}) \Rightarrow [x_{t-1} + \epsilon]$

$\sim \mathcal{N}\left(x_t \,\middle|\, \mu_t = \sqrt{1-\beta_t} * x_{t-1};\ \Sigma_t = \sqrt{\beta_t}\, I\right)$

$\beta_t \Rightarrow$ very close to zero

$1-\beta_t \Rightarrow$ close to 1

$$\boxed{x_t = \sqrt{1-\beta_t} * x_{t-1} + \sqrt{\beta_t} * \epsilon}$$



$x_0 \quad x_1 \quad x_2 \quad x_3 \quad x_{50}$ — x_T

forward

x_t sampled from $q(x_t | x_{t-1})$

$$q(x_t|x_{t-1}) \sim \mathcal{N}\left(x_t \,\middle|\, \mu_t = \sqrt{1-\beta_t} * x_{t-1}\,;\ \Sigma_t = \sqrt{\beta_t}\, I\right)$$

$$\boxed{x_t = \sqrt{1-\beta_t}\; x_{t-1} + \sqrt{\beta_t} * \epsilon_{t-1}}$$

$$[x_0] \rightarrow [x_1] \rightarrow [x_2] \rightarrow [x_3] \rightarrow [x_{50}]$$

$$[x_t \longleftarrow x_0] \text{ in } (1 \text{ step})$$

$$\alpha_t = 1 - \beta_t \rightsquigarrow [\beta_t = 1 - \alpha_t]$$

$$\tilde{\alpha}_t = \prod_{s=0}^{t} \alpha_s$$

(*) $\epsilon_{t-1}, \epsilon_{t-2}, \epsilon_{t-3} \ldots\ldots \Rightarrow \epsilon$ (WIG)

(C) $x_t = \sqrt{1-\beta_t} * x_{t-1} + \sqrt{\beta_t} * \epsilon_{t-1}$

(D) $x_t = \sqrt{\alpha_t} * x_{t-1} + \sqrt{1-\alpha_t}\; \epsilon_{t-1}$

(E) $x_{t-1} = \sqrt{\alpha_{t-1}} * x_{t-2} + \sqrt{1-\alpha_{t-1}} * \epsilon_{t-2}$

(F) $x_t = \left\{ \underbrace{\sqrt{\alpha_t \alpha_{t-1}} * x_{t-2}} + \underbrace{\sqrt{\alpha_t - \alpha_t \cdot \alpha_{t-1}} * \epsilon_{t-2}} \right\}$ $\hookrightarrow$ (X) $\sim \mathcal{N}(0, \sqrt{\alpha_t - \alpha_t \alpha_{t-1}})$

$+ \underbrace{\sqrt{1-\alpha_t} * \epsilon_{t-1}}$

[Detour]

(Y) $\sim \mathcal{N}(0, \sqrt{1-\alpha_t})$

$X \sim \mathcal{N}(\mu_x, \underset{SD}{\sigma_x}) \Rightarrow (\sigma_x)^2$

$Y \sim \mathcal{N}(\mu_y, \underset{SD}{\sigma_y}) \Rightarrow (\sigma_y)^2$

$[Z = X + Y]$ if $Z \sim \mathcal{N}(\mu_x + \mu_y ; \underset{SD}{\sqrt{\sigma_x^2 + \sigma_y^2}}) \rightsquigarrow (\sigma_x^2 + \sigma_y^2)$

$Z \Rightarrow (0, \sqrt{\alpha_t - \alpha_t \cdot \alpha_{t-1} + 1 - \alpha_t})$

$\Rightarrow (0, \sqrt{1 - \alpha_t \alpha_{t-1}})$

(G) $x_t = \sqrt{\alpha_t \alpha_{t-1}} * x_{t-2} + \sqrt{1-\alpha_t \alpha_{t-1}} * \epsilon$

(H) $x_t = \sqrt{\underbrace{\alpha_t \alpha_{t-1} \alpha_{t-2} \cdots}} * x_0 + \sqrt{1 - \alpha_t \alpha_{t-1} \cdots \alpha_1} * \epsilon$

$x_t = (\sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t} \cdot \epsilon)$

$\hookrightarrow \sim \mathcal{N}(\sqrt{\bar{\alpha}_t}\, x_0 , \sqrt{1-\bar{\alpha}_t} \cdot I)$

$$x_t \sim \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\, x_0,\ \sqrt{1-\bar{\alpha}_t} * I\right)$$

Schedule for β

[α depends on β] β is fixed

$\beta \Rightarrow$

10^{-4}	$10^{\cdots}$	- -	-	-	-	-	-	-	-	.	.	0.2

β_0 β_1 β_2 β_3 - - - - - β_T

$$q(x_t \mid x_{t-1})$$

$$q(x_{t-1} \mid x_t)$$

$$\Downarrow$$

$$q(x_{t-1} \mid x_t, t) \rightarrow q(x_{t-1} \mid x_t, x_0)$$